

Movement of Solutes in Tropical Alfisols Cleared by Different Methods

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ABSTRACT

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Solute movement under saturated water-flow conditions was studied in undisturbed and disturbed soil columns and in the field under four methods of clearing: manual; shear blade; tree pusher; and root rake, and traditional farming. Soil under forest was included as a control.

Chloride (Cl^-) appeared in the effluent at 0.1 pore volume after its application to the surface of undisturbed soil columns taken from plots cleared manually, with shear blade, or with tree pusher. In columns from forest and traditional farming plots, no Cl^- appeared in the effluent until 0.2 pore volume of solution had passed through the columns. The pore volume required to attain 0.5 relative Cl^- concentration was 0.60, 0.60, 0.40, 0.40, and 0.26 for the columns from forest control, traditional farming, shear-blade, tree-pusher and manually cleared plots, respectively. This indicates preferential water flow in the columns from manually cleared plots as compared to others. Nitrate leaching in the undisturbed columns was nearly identical to chloride. Phosphorus appeared in the effluent after the passage of 0.34 pore volume of solution through the columns of forest control and manually cleared treatments, and of 0.7 pore volume in the shear blade treatment. The increase in the relative phosphate concentration was considerably slower than chloride and nitrate. Chloride appeared later in the repacked columns than in the undisturbed columns, indicating the absence of preferential flow pathways in the disturbed columns.

In the field, less than 100% Cl^- and water were retained in the 0-70 cm depth, indicating some flow of chloride solution to below 70 cm depth.

INTRODUCTION

Solute movement by miscible displacement (Nielsen and Biggar, 1961) is an important process because of its effect on nutrient and chemical displacement in the soil. Previous studies have shown the dependence of miscible displacement efficiency of non-interacting solutes on pore water velocity (Biggar and Nielsen, 1967), initial soil water content particularly in the vicinity of the soil surface (Nielsen and Biggar, 1961; Warrick et al., 1971; Ghuman et al., 1975), anion exclusion (Kemper, 1960), pore size (Kanchanasut et al., 1978),

aggregate size (Passioura and Rose, 1971), soil layering (Selim et al., 1977; Ghuman and Prihar, 1980a), soil type (Ghuman and Prihar, 1980b) and evaporation rate (Ghuman and Prihar, 1983). Boast (1973) reviewed mathematical models used by different researchers to predict simultaneous transport of salt and water in soils.

In the humid and subhumid tropics, large areas of forested lands are annually being cleared for agricultural purposes. According to Johnson (1981), the world's forests are disappearing at the rate of 20 million ha per year, of which about 11 million ha are in the equatorial wet tropics. Land clearing is achieved by different methods, manual, mechanical or chemical, depending on the resources available, intended land use, and the size of the holding. In general, mechanical methods of land clearing and development cause more soil compaction and displacement of surface soil (Seubert et al., 1977; Lal and Cummings, 1979), and more soil erosion (Lal, 1981) than the manual methods. The magnitude of the effects of land-clearing methods on soil physical and nutritional properties depends on soil characteristics, antecedent moisture content, vegetation, and the method used (IITA, 1982).

There is little available information regarding the effects of different land-clearing methods on water and nutrient movement in soils of the tropics. This type of research information is necessary for developing agronomic practices that facilitate efficient utilization of applied nutrients, and also is relevant for understanding and controlling pollution hazards and eutrophication of natural waters. It is for these reasons we investigated anion movement under saturated-flow conditions in undisturbed and disturbed soil columns in the laboratory and in fields cleared by four land-clearing techniques.

MATERIALS AND METHODS

A 44-ha watershed, subdivided into 12 plots of 3 to 4 ha each, was cleared from forest in April 1979. Land clearing methods included: 1) manual; 2) shear blade; 3) tree pusher and root rake; and 4) traditional farming. Uncleared forest soil was considered as control. The traditional farming technique involves slashing of shrubby undergrowth and felling of trees which are burnt later, when dry; furthermore, the stumps and roots of trees are not eradicated and large trees are left standing (Webster and Wilson, 1980). The field layout and clearing treatments have been described in detail elsewhere (Lal, 1981). Prior to deforestation, the watershed had been under secondary forest regrowth for more than 15 years (Couper et al., 1981). The soils belong to the Egbeda/Iwo association on the upper part of the catena (clayey skeletal, kaolinitic, isohyperthermic, Oxic Paleustalf) and to the Apomu series (Psammentic Ustorthent) towards the valley bottom (Moormann et al., 1975). The texture of the surface layer of the Egbeda/Iwo association at the sampling site is sandy clay loam changing into sandy clay at 20 cm depth and clay at 60 cm depth. Coarse

TABLE 1

Representative properties of the soils studied

Land clearing method	Soil layer (cm)	Composition (%)				pH (1:2 soil:water)	Organic C (g kg^{-1})	Bulk density (Mg m^{-3})
		Sand	Silt	Clay	Gravel			
Forest	0-5	61	16	23	1	5.6	30	0.98
	5-10	65	14	21	20	5.9	10	1.39
	10-18	65	12	23	37	5.6	8	1.52
Manual	0-5	57	20	23	2	6.3	29	1.10
	5-10	61	16	23	5	6.1	22	1.41
	10-18	61	12	27	34	6.5	13	1.52
Shear blade	0-5	63	16	21	3	5.1	14	1.44
	5-10	63	12	25	32	5.8	9	1.47
	10-18	59	12	29	60	5.5	6	1.58
Tree pusher	0-5	61	12	27	30	5.9	19	1.26
	5-10	59	14	27	17	6.0	11	1.55
	10-18	53	14	33	45	6.0	9	1.72
Traditional farming	0-5	47	20	33	1	6.1	27	1.02
	5-10	49	16	35	2	6.5	21	1.14
	10-18	49	14	37	6	6.2	15	1.33

material ($> 2 \text{ mm}$) of different sizes is present in the profile, with gravel content ranging from 3 to 80%. The pH of the surface soil ranges from 5.1 to 6.5. Soil under forest contains up to 20 g kg^{-1} organic C in the 0-5-cm layer. Post-clearing management practices in some of the plots included no-tillage and conventional tillage as described by Couper et al. (1981). However, the results presented in this report pertain to no-tillage plots only.

LABORATORY EXPERIMENTS

Undisturbed columns

Plexiglass columns of 21 cm length and 10 cm diameter were used to obtain undisturbed soil cores from the top 18 cm of different plots. Particle size distribution, pH, organic carbon and bulk density of the soils sampled are given in Table 1. The columns were taken during the rainy season (April-September, 1982) when the soil was moist, so as to facilitate sampling. The sampling sites for column extraction were selected at random on the upper part of the plots.

Column sampling procedure

When the soil was at or near field capacity, a column was pushed vertically into the soil to about 5 cm depth. The surrounding soil was gradually excavated to a depth of 20 cm. The soil column thus formed was slightly larger in diameter than the plexiglass encasing, which was gradually slipped down over the column in 1–2-cm increments. If the column advance was inhibited by particles of gravel, these were carefully removed from the soil body and the gaps left were filled with a sand-cement paste before pushing the column any further. When the casing reached the desired depth, the column was removed by breaking off the core from the main soil body. The superfluous soil was removed from the base by breaking the aggregates along the natural cleavage lines to prevent smearing and blockage of conducting pores. The base of the core was covered with a piece of cloth held in place by a rubber band. In this way, four soil columns were collected from each of the land-clearing treatments and from the control plots. The soil cores were then transported to the laboratory where they were allowed to dry at room temperature to some undefined moisture content for at least two days. The drying process hardened the sand-cement paste into artificial gravel. This procedure was necessary to obtain an undisturbed soil core from a gravelly profile.

A. Chloride leaching. Two columns per treatment were saturated by capillarity, after immersing each in a beaker containing 0.005M CaSO_4 solution. Saturated columns were then carefully placed in a Buchner funnel fitted in a stand, and leached again with the saturation solution (0.005M CaSO_4) by maintaining a head of approximately 2 cm on the soil surface. When constant flux was achieved, the supply of this solution was discontinued. When the ponded solution had just soaked into the soil from the surface, it was replaced by 0.05M CaCl_2 solution. The effluent was collected in 30-ml aliquots which were analysed for chloride with standard AgNO_3 using 5% KCrO_4 as an indicator (Richards, 1968).

B. Nitrate and phosphate leaching. Another set of columns (two per treatment) was saturated with 0.003M KCl solution and then leached. After reaching steady-state flow condition, the columns were leached with a solution containing 0.001M KNO_3 and 6.5×10^{-4} M P as KH_2PO_4 . The potassium chloride concentration in the displaced solution was chosen so that the displacing solutions had the same density, to avoid gravity segregation effects. The leachate, collected in 30-ml aliquots, was analyzed for nitrate and phosphate with an auto-analyzer.

Disturbed columns

Soil samples were collected at 0–5, 5–10 and 10–18-cm depths from different land-clearing treatments and control plots, and air dried. Soil samples from a given layer were divided into two portions. The gravel was removed from one set of samples by sieving through a 2-mm sieve. The second set, however, was not sieved and all the gravel remained in the soil.

Columns were packed in duplicate with the disturbed soil at bulk densities nearly similar to that of undisturbed columns (Table 1). The columns were then saturated and leached with solution as described above.

Field experiment

The field experiment was conducted at a site adjacent to where the core samples were obtained (the upper part of the respective plots). A maize crop was growing in the cleared plots at the time of this experiment. Plants were cut flush with the ground before making a subplot in a given treatment. A wooden 1.45×1.45 -m frame of 0.30-m depth was used to demarcate the subplot. The frame was installed 15 cm below and 15 cm above the soil surface to restrict lateral water flow near the soil surface, and runoff, respectively. This subplot was irrigated with 12–15 cm of irrigation water by keeping nearly a 2-cm head on the surface. Subsequent to the infiltration of irrigation water, 5 cm of 0.01 M CaCl_2 solution was ponded on the soil surface. Immediately after infiltration of the CaCl_2 solution, soil samples from three random locations in the subplot were obtained with a soil auger in 10-cm increments from 0 to 70 cm depth. Samples for the same depths were composited. After oven-drying at 105°C for 24 h, samples were milled and water extracts from 1:2 soil water suspensions were collected. Extracts were analyzed for Cl^- with standard AgNO_3 solution (Richards, 1968).

Presentation of results

Leaching results are presented in the form of breakthrough curves with relative anion concentration as the ordinate and pore volume of effluent as the abscissa. The relative anion concentration (c/c_0) was calculated from the ratio of anion concentration in the effluent (c) to that in the influent (c_0). The pore volume at any instant was determined as follows:

$$\text{Pore volume} = \frac{\text{Cumulative volume of effluent collected}}{\text{Volume of water in the soil column}}$$

The volume of water in the soil column was determined by oven-drying the soil at 105°C after the completion of the leaching run.

TABLE 2

The differences in chloride leaching patterns between the two replicates (R_1 and R_2) collected at each treatmental site

Land-clearing method	Pore volume to reach c/c_0 :					
	0.5			0.9		
	R_1	R_2	Diff.	R_1	R_2	Diff.
Forest	0.58	0.56	0.04	1.46	1.49	0.03
Manual	0.26	0.25	0.01	1.47	1.49	0.02
Shear blade	0.38	0.45	0.07	1.90	1.90	0.00
Tree pusher	0.37	0.46	0.09	1.82	1.85	0.03
Traditional farming	0.59	0.60	0.01	1.61	1.65	0.04

RESULTS AND DISCUSSION

Undisturbed soil columns

A. Chloride. The leaching characteristics of chloride from the two replicated columns taken at a given site were reproducible (Table 2). To reach $0.5 c/c_0$ between the two replicates, the difference in pore volumes was greatest for the tree-pusher and shear-blade treatments and least for the manual and traditional farming. However, the pore-volume differences among the replicates were smaller at $0.9 c/c_0$. The larger differences in pore volumes at $0.5 c/c_0$ were caused by surface disturbances associated with the heavy machines used in the shear-blade and tree-pusher treatments. As a result of small differences between the two samples, the breakthrough curves presented hereafter are the averages of these duplicate columns.

Methods of land clearing had measurable effects on chloride movement through soil columns (Fig. 1). Chloride appeared in the effluent almost immediately (at 0.1 pore volume) after application to the surface of columns sampled from plots cleared either manually, with shear blade, or with tree pusher. But in columns from plots under forest and traditional farming, Cl^- did not appear in the effluent until 0.2 pore volume of solution had passed. Relative Cl^- concentration increased rapidly after appearance to the maximum (0.94) in the columns obtained from plots cleared manually, followed by shear blade or tree pusher, then by forest control or traditional farming. The pore-volume equivalent to attain 0.5 relative Cl^- concentration also varied among treatments and was 0.60, 0.60, 0.40, 0.40 and 0.26 for the forest control, traditional farming, shear blade, tree pusher and manual clearing treatments, respectively. Water movement and hence solute movement is controlled by soil struc-

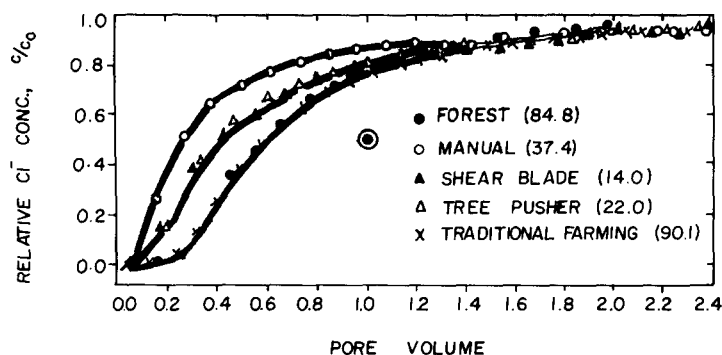


Fig. 1. Chloride breakthrough curves for undisturbed soil cores taken from plots cleared by four methods. The numbers in the legend are the steady-state flow rates in cm/h. The circled symbol indicates $c/c_0 = 0.5$ at one pore volume.

ture (i.e., pore size distribution) under saturated flow and mainly by texture under unsaturated flow conditions (Hillel, 1971). Though the clay content at the traditional farming site was relatively higher than at the other site (Table 1), the pore volume differences in the arrival of the anion at the exit end of columns, and consequently in the position of the breakthrough curves for different treatments, were due to the land-clearing treatments affecting the saturated flow.

The shape or position of breakthrough curves with respect to the point of 0.5 relative chloride concentration and 1 pore volume is indicative of preferential flow (Fig. 1). The slow rise to $0.94 c/c_0$ shows that not all columns from all the treatments contained continuous soil pores. Had the pores been continuous, relative concentration in the effluent should have risen immediately after the surface application of chloride solution (Kanchanasut et al., 1978). Soils from the forest and traditional farming treatments were relatively more aggregated than those from plots cleared by manual, shear-blade and tree-pusher methods (Lal et al., 1980). Forest soil (and also that farmed traditionally) contained approximately equal proportions of smaller and larger pores. In shear-blade and tree-pusher clearing treatments, however, it seems that a majority of the larger pores in the surface layers disappeared due to the crushing action of the clearing equipment, thus rendering small-sized pores predominant and destroying continuity of transmission pores.

The differential pore-size distribution among soil columns from different treatments was also substantiated by differences in steady flux rates shown in Fig. 1. Though fluxes were higher in columns sampled from the forest and traditional farming plots than in those from other treatments, their subsequent rise to the maximum relative concentration was comparatively slow. This was perhaps caused by the more uniform advance of the Cl^- front through well-distributed small and large pores in forest and traditionally farmed soils.

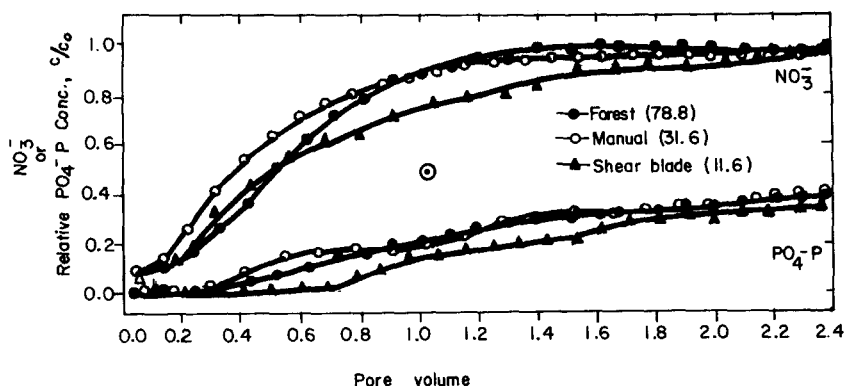


Fig. 2. Nitrate and phosphate breakthrough curves for undisturbed soil cores taken from plots cleared by two methods. The numbers in the legend are the steady-state flow rates in cm/h. The circled symbol indicates $c/c_0 = 0.5$ at one pore volume.

However, in manually cleared soil there might have been some large channels left by decayed roots of burned secondary regrowth of forest, thereby causing more preferential flow than in other clearing treatments. Preferential transport of anions through undisturbed soil cores has also been reported by Scotter and Kanchanasut (1981) and substantiated by Smettem and Trudgill (1983) using dyes.

B. Nitrate and phosphate. Since shear blade and tree pusher or forest and traditional farming treatments had similar Cl^- breakthrough curves, and NO_3^- movement is generally similar to chloride under saturated conditions (Corey et al., 1963), data for forest, manual, and shear-blade clearing treatments only are presented in Fig. 2. In the initial stages of the leaching process, the NO_3^- movement as affected by different land clearing methods was similar to that of the Cl^- . At 0.5 pore volume, the breakthrough curve for the forest control crossed that of the shear-blade treatment and thus attained the maximum concentration noted earlier. At 2.4 pore volume, there was not much difference in the relative NO_3^- concentration among columns sampled from different treatments.

Phosphorus appeared in the effluent only when 0.34 pore volume of solution had passed through the columns of forest control and manually cleared treatments (Fig. 2). In columns from the shear-blade clearing treatment, P appeared after the passage of 0.7 pore volume. These results are not in agreement with those of Scotter and Kanchanasut (1981) who reported that P appeared immediately after its application to the soil surface of Tokomaru silt-loam columns. This disagreement could be caused by pore-size distribution in general and pore continuity in particular, or by differences in the retention (adsorption)

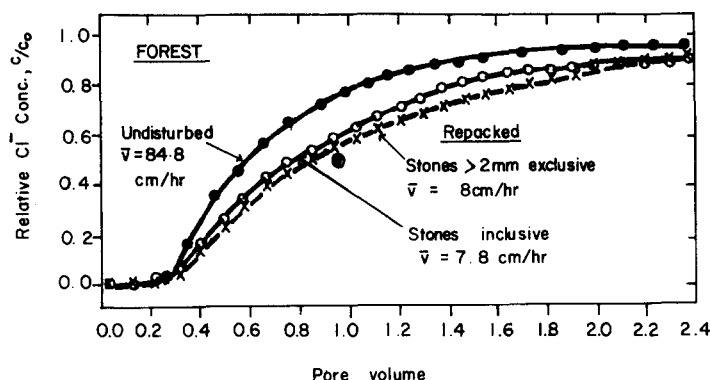


Fig. 3. Comparison of chloride breakthrough curves for repacked and undisturbed soil cores taken from forest control. $\bar{V}$ is steady-state flow rate and circled symbol for $c/c_0 = 0.5$ at one pore volume.

capacity of the soils involved. Data in Fig. 2 further show that the increase in relative phosphate concentration was slower than that of the NO_3^- . This difference between the NO_3^- and phosphate data indicates that adsorption had a measurable effect which could be caused by diffusion and adsorption of P directly from the ponded solution before it enters the soil (Kanchanasut et al., 1978), and adsorption during mass flow through the soil.

Undisturbed vs. repacked columns

The repacked soil columns from the forest control yielded different breakthrough curves than did the undisturbed columns from the same treatment (Fig. 3). Chloride appeared in the effluent slightly later in repacked than in undisturbed columns, particularly when the soils were devoid of gravel > 2 mm size. Similar results were reported by Elrick and French (1966) and Kissel et al. (1973). The relative Cl^- concentration reached 0.5 at 0.76 and 0.84 pore volume in repacked columns with and without gravel, respectively, rather than at 0.60 pore volume as in undisturbed columns. These results show that displacement of Cl^- in repacked columns was relatively more complete than in undisturbed columns, probably because of the absence of preferential pathways. The absence of larger pores or preferential pathways in unaggregated, repacked columns was also indicated by flow velocities that were about 10.5 times lower than in undisturbed columns. The breakthrough curves for repacked columns are comparable with those of Nielsen and Biggar (1961).

Chloride breakthrough curves from repacked columns for other treatments closely resembled those from the forest control. For example, relative chloride concentration of 0.5 was attained at 0.92, 0.84, 0.84, and 0.80 pore volume, for columns reconstituted with soil from the traditional farming, shear-blade,

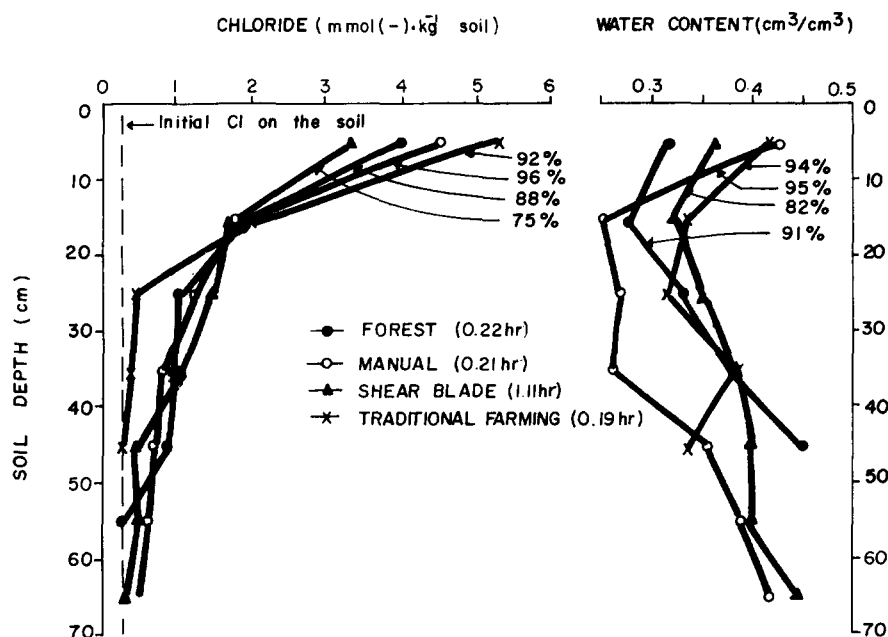


Fig. 4. Chloride and water distribution with depth immediately after infiltration of 5 cm of 0.01 M CaCl_2 solution in the field plots cleared by three methods. The bracketed number in the legend is infiltration time taken by 5 cm of solution for a given treatment.

manual, and tree-pusher clearing treatments, respectively; the respective steady-state flow velocities were 3.9, 3.0, 4.0, and 3.2 cm h^{-1} .

Field experiments

As in the laboratory studies, the field experiments also showed some preferential flow as indicated by less than 100% retention of the applied Cl^- and water in the top 70 cm depth immediately after infiltration ceased (Fig. 4). Kolenbrander (1970), Kissel et al. (1973) and Wild and Babiker (1976) also reported similar results. Furthermore, the least Cl^- retention (75%) was observed for the soil in shear-blade plots, while the greatest (96%) was recorded in the manually cleared plots. The lowest level of retention in the shear-blade plots was attributed to more leaching losses beyond 70 cm depth, due to increased infiltration time taken by 5-cm salt solution (also shown in Fig. 4). The average flow rates, which were calculated on the basis of time taken by 5 cm of salt solution to infiltrate into the soil, were 22.8, 4.5, 23.1 and 26.1 cm/h in forest, shear-blade, manual and traditional farming treatments, respectively. These flow rates are 1.5–4 times lower than those observed in undisturbed columns.

CONCLUSIONS

The nonreactive solutes (Cl^- and NO_3^-) in the effluent from soil columns collected from plots cleared manually and with shear blade (at 0.1 pore volume) appeared earlier than from forest control and traditional farming (0.2 pore volume). The reactive solute (phosphate) appeared earlier in the leachate from columns of forest control and manual treatments (0.34 pore volume) than from the shear blade (0.7 pore volume). Phosphate appeared in the effluent much later than the nonreactive Cl^- and NO_3^- . Furthermore, Cl^- in the effluent from undisturbed columns appeared earlier than from disturbed. This shows the importance of conducting salt movement studies for leaching characteristics of soils on undisturbed soil cores which are more representative of field conditions. Field studies showed some preferential solution flow.

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